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latihan soal Resistor

1). Berapakah nilai tahanan untuk resistor :

a. kuning, ungu, emas, perak
4,7 Ω ± 10 %

b. orange, putih, kuning, emas
~~39k Ω ± 5%~~ 390k Ω ± 5%

c. cokelat, hitam, merah, emas
1000 Ω ± 5%

d. coklat, hijau, hijau
1500k Ω

e. hijau, biru, hitam, merah, coklat
56k Ω ± 1%

F. coklat, abu, hitam, hitam, coklat
180Ω ± 1%

2). Hitunglah daya pada masing masing tahanan :

a. 100mA → [100 Ω]

$P = I^2 . R$

$= (0,1)^2 . 100 = 1$ watt

b. → [1k Ω]

← 5V →

$P = \frac{V^2}{R} = \frac{5^2}{1k} = 0,025$ watt

c. 1A → [R]

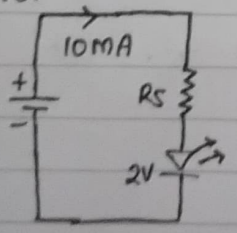
← 5V →

$P = V . I$

$= 5 . 1 = 5$ watt

3). Tentukan nilai resistor pada rangkaian dibawah ini dgn nilai standar resistor

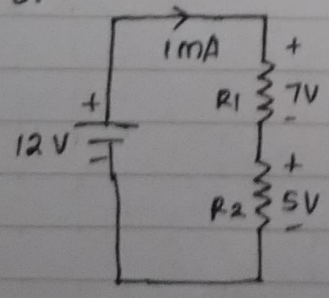
a.



$I = 10mA = 0,01$ A

$R = \frac{V}{I} = \frac{5}{0,01} = 500\ \Omega$

b.

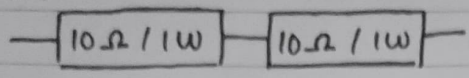


$I = 1mA = 0,001$ A

$R = \frac{V}{I} = \frac{12}{0,001} = 12k\ \Omega$

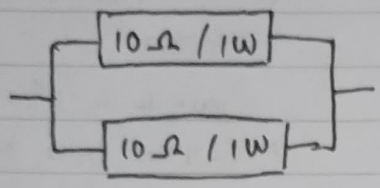
4). Hitung besar tahanan total bila resistor 10Ω / 1W di susun secara :

a. seri



$R_{seri} = R_1 + R_2 = 10 + 10 = 20\Omega$

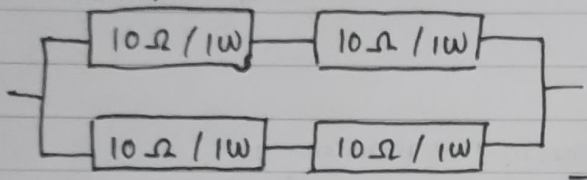
b. paralel



$$\frac{1}{R_{paralel}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$= \frac{1}{10} + \frac{1}{10} = \frac{1+1}{10} = \frac{2}{10} = \frac{1}{5} = 5\Omega$$

c. Seri paralel



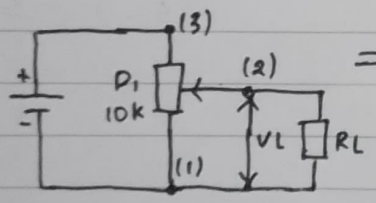
$R_S 1 = 10 + 10 = 20\Omega$

$R_S 2 = 10 + 10 = 20\Omega$

$$\frac{R1}{R_{total}} = \frac{1}{R_{S1}} + \frac{1}{R_{S2}} = \frac{1}{20} + \frac{1}{20} = \frac{1}{10} = 10\Omega$$

5). Gambarkan grafik $V_L = f(P_{1-2})$ untuk kondisi :

a). $R_L = \infty$



$\Rightarrow$

$V_L = I\,(P // R_L)$

$= 0(0) = 0V$

$$I = \frac{\mathcal{E}}{P_1 // R_L} = \frac{12}{10k / \infty} = \frac{12}{0}$$

$R_a \rightarrow R_b = P_1 = 10k$

$R_b = P_1 - 2$

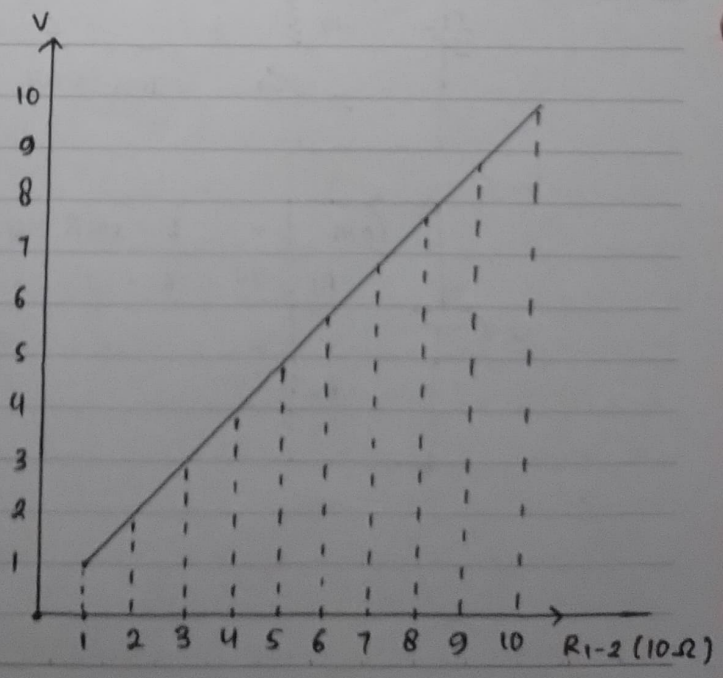
R_b nilainya 0, 1, ... 10k

$$V_L = \frac{R_L // R_b}{(R_L // R_b)} \times 10V$$

$$= \frac{R_b}{R_b + R_a} \times 10$$

$R_b = 0 \quad R_a = 10$

$$V_L = \frac{0}{0+10} \times 10 = 0$$



No.

b. $R_L = 5k$

$$V_L = \frac{5k \,//\, R_b}{(5k \,//\, R_b + R_a)} \times 10$$

$R_b = 0$

$$V_L = \frac{5k \,//\, 0}{(5k \,//\, 0^2 + 0 + 10)} \times 10 = 0 \text{ volt}$$

$R_b = 0$

$$V_L = \frac{5k \,//\, 1k}{5k \,//\, 11k + 9k}$$

$R_b = 0$

$$V_L = \frac{5k \,//\, 0}{(5k \,//\, 0 + 10)} \times 10 = 0 \text{ VoH}$$

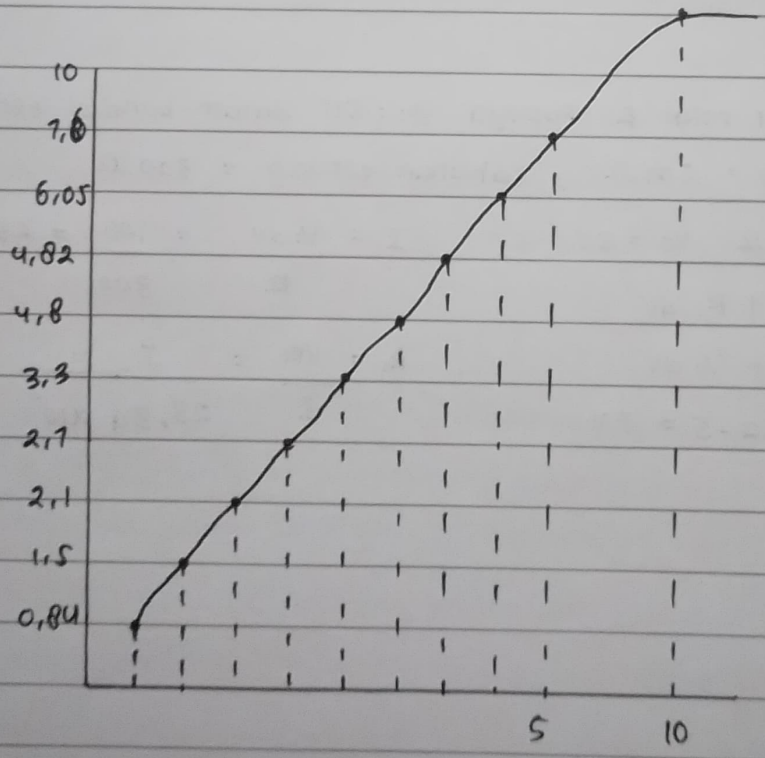
→ $R_b = 1k$

$$V_L = \frac{5k \,//\, 1k}{(5k \,//\, 1k + 9k)} \times 10$$

$R_b = 2k$

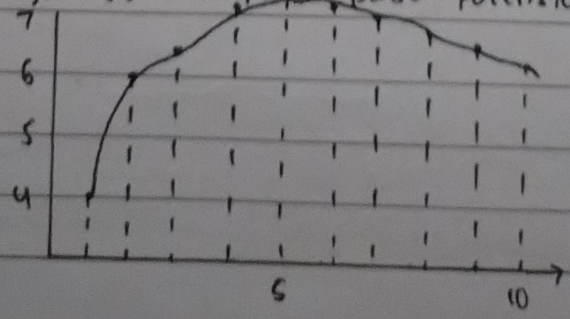
$$V_L = \frac{5k \,//\, 2k}{(5k \,//\, 2k + 9k)} \times 10$$

$= 1{,}5\ V$



6). Nilai resistor variabel berubah dari 0-10 Ω (P_{1-2}). Gambar diagram yg sama

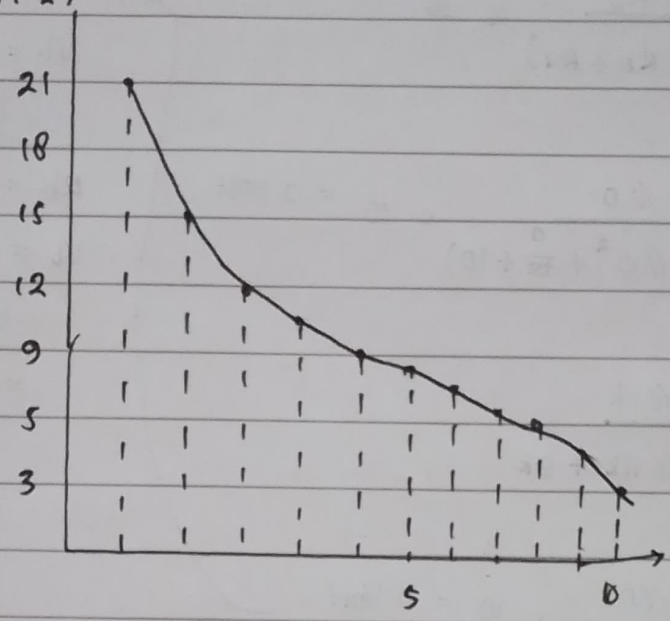
a. Grafik daya yg disisipkan pada potensiometer $P(pot) = f(P_{1-2})$



80 mm x 257 mm
KIKY

No. :

b. Grafik daya yg disipasikan pada tahanan dalam batre $P(R_1) = F(F(P_1-2))$



7). Tentukan besar nilai R_1 supaya $V_0 = 5V$ dalam kondisi terang.

Tahanan gelap = $2M\Omega$, tahanan terang = 300Ω

↳ $R = 300\Omega$ $V_0 = 5V$

$V_2 = V_{R1} + R_1 dV$

$12V = 5 + V_1 dV$

$V_1 dV = 12 - 5 = 7V$

$I = \frac{V_1 dV}{R} = \frac{7.00}{300} = 23,34\ mA$

$R_1 = \frac{V_{e1}}{I} = \frac{I}{23,34 \times 10^{-3}} = 214\ \Omega$